FYTA — Data Engineer assessment

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Three weeks, 8 soil sensors, 15,950 readings, 49 user notes, 28 plant photos. `uv run fyta_pipeline.py` reproduces every number below (`out/run.log`); `uv run test_gates.py` runs 18 regression checks, each verified to fail when its fix is reverted; `fyta_analysis.ipynb` executes from a fresh kernel.

Findings

SENS-06 moisture stayed at 25.01 for 1,344 readings — fourteen days — while its other four channels kept moving. I nulled that interval and kept a reason per row. The cause is unresolved.

SENS-07 reads Fahrenheit and runs two hours off local time. Its temperature scale and offset fit the Fahrenheit conversion, and aligning its light series against the fleet supports a two-hour clock offset (r 0.802 → 0.9997). Correlation alone sees neither: it correlates at 0.9843 with its peers while in the wrong unit.

SENS-05 missed 193 slots (48.5 h) with no empty cells anywhere — the rows are absent, so only an expected-slot count finds them. **SENS-08 delivered 15 duplicate rows** with identical payloads, so deduplicating them is lossless. **Plant IDs repeat:** joining logs or photos to the device map expands 49 rows to 67 and 28 to 35.

Two questions stay open: SENS-03's EC climbs in a straight line from 06-11 with no logged cause, and the two-sensor plants disagree (r −0.08, +0.14), which depth, placement or a failed probe would each explain.

Treatment and outputs

Raw columns are never edited. Corrections land in `_clean` columns beside them: physically impossible values are nulled and recorded in `out/qc_rejections.csv`, while values merely outside the published range are kept and flagged — 21% of SENS-04's readings sit above the 65% ceiling, and that excursion is the shape of its drying cycle. Nothing is clipped, averaged or interpolated.

`out/plant_features.csv` (11,903 rows, grain `(user_plant_id, ts_aligned)`, asserted) carries the sensor channels and plant metadata. It excludes logs and photos: spreading 49 events and 28 photos across 15-minute rows would duplicate a sparse event hundreds of times. Both probes are kept, with `vwc_delta` between them.

Cross-source result

The image model's water-deficiency score tracks sensor dryness at r = 0.75 — the axis the sensor already measures — and one synthetic generator could have produced both, so independence is unestablished. Its other heads never exceed 0.06.

For the five fertilising logs, the selected windows established no corroboration: screening five events at 3σ against each probe's own noise is low-power, and the log is not a known event time. A wider window would have made UP-1002 look confirmed, from SENS-03's drift rather than chemistry.

Limits and next steps

Completeness is measured against each device's observed span, not an installation window; grid spacing and phase are asserted, boundary slots are not. The frozen-channel rule misses a light sensor that dies at exactly zero, and SENS-03's drift is not flagged per row. Unit fixes, clock offsets and ranges rest on published specs and the substrate conversion curve, which FYTA can confirm (D1, D7).

For production: one compacted Parquet object per device per day ($15/month of requests at 100k devices), with the per-device-window checks running hourly off the ingestion buffer — hourly lake files cost 24× that to hold four readings each. Arithmetic in `docs/PIPELINE_DESIGN.md` §5.